1 Oct 20 HW

Oct 20 homework: Proov 3 equation

1.$[p_x, \psi(x)] = -i\bar{h}\frac{\partial\psi}{\partial x}$

2.$[p_x^2, \psi(x)] = -2i\bar{h}\frac{\partial\psi}{\partial x}p_x - \bar{h}^2\frac{\partial^2\psi}{\partial x^2}$

3.$[x, y(p(x))] = i\bar{h}\frac{\partial y(p(x))}{\partial p(x)}$

proov now:

1:$[p_x, \psi]\psi = p_x[\psi^2] - \psi p_x[\psi] = -i\bar{h}2\psi\partial_x\psi + i\bar{h}\psi\partial_x\psi = -i\bar{h}\partial_x\psi\psi$

2:$[p_x^2, \psi] = -\bar{h}^2\partial_x\partial_x\psi^2 + \bar{h}^2\psi\partial_x\partial_x\psi = -2\bar{h}i\partial_x\psi \times p_x\psi - \bar{h}^2\psi\partial_x\partial_x\psi = (-2\bar{h}i\partial_x\psi p_x - \bar{h}^2\partial_x\partial_x\psi)\psi$

3:$[x, y(p(x))]$ and$y(p(x)) = \Sigma_n C_n p(x)^n$ so $\Sigma_n C_n[x, p_x^n] = [x, y(p(x))]$ and $[x, p_x^n]\psi = x \times p_x^n\psi - p_x^n(x\psi) = in\bar{h}p_x^{n-1}\psi = \Sigma_n C_n[x, p_x^n] = [x, \Sigma_n c_n p_x^n] = i\bar{h}\Sigma_n C_n n p_x^{n-1}\psi = i\bar{h}\partial_{p_x}y(p_x)\psi$

$p_x^n(x\psi) = p_x^{n-1}(c\psi + x\partial_x\psi) = cp_x^{n-1}\psi + p_x^{n-2}(c\partial\psi + xp_x^2\psi)$ so $= ncp_x^{n-1}\psi + p_x^n\psi$and$c = -i\bar{h}$

$p_x[x\psi] = -i\bar{h}\psi + xp_x\psi$

D O N E .

2 Oct 23HW part2

PROOF:$e^{a\frac{d}{dt}}\psi = \psi(x+a)$

$e^{a\frac{d}{dt}}\psi = \Sigma_n \frac{a^n}{n!}\frac{d^n}{dt^n}\psi; \psi(x+y) = \Sigma_n \frac{\frac{d^n\psi}{dt^n}|_{y=0}}{n!}y^n$consider y=a:$\Sigma_n \frac{\frac{d^n\psi(x)}{dt^n}}{n!}a^n = \Sigma_n \frac{a^n}{n!}\frac{d^n}{dt^n} = e^{a\frac{d}{dt}}\psi$

DONE.

3 Oct 28HW

A:

let $l_\pm = l_x \pm il_y$

PROOF:

(a).$[l_z, l_\pm] = \pm\bar{h}l_\pm$

(b).$l_\pm l_\mp = l^2 - l_z^2 \pm \bar{h}l_z$

(c).$[l_+, l_-] = 2\bar{h}l_z$

proof a:

$[l_z, l_\pm] = [l_z, l_x] \pm i[l_z, l_y] = i\bar{h}l_y \pm i(-i\bar{h}l_x) = i\bar{h}l_y \pm \bar{h}l_x = \pm\bar{h}l_\pm$

proof b:

$l_\pm l_\mp = (l_x \pm il_y)(l_x \mp il_y) = l_x^2 \mp il_xl_y \pm il_yl_x + l_y^2 = l_x^2 + l_y^2 \pm i(l_yl_x - l_xl_y) = l^2 - l_z^2 \pm i[l_y, l_x] = l^2 - l_z^2 \pm \bar{h}l_z$

proof c:

$l_+l_- - l_-l_+ = (l_x + il_y)(l_x - il_y) - (l_x - il_y)(l_x + il_y) = -il_xl_y + il_yl_x - il_xl_y + il_yl_x = 2i[l_y, l_x] = 2\bar{h}l_z$

B:

PROOF:

(a) $p \times l + l \times p = 2i\bar{h}p$

(b)$i\bar{h}(p \times l - l \times p) = [l^2, p]$

proof a:

$(p_1l_2 - p_2l_1)k + (p_2l_3 - p_3l_2)i + (p_3l_1 - p_1l_3)j + (l_1p_2 - l_2p_1)k + (l_2p_3 - l_3p_2)i + (l_3p_1 - l_1p_3)j =$

$([p_1, l_2] + [l_1, p_2])k + ([p_2, l_3] + [l_2, p_3])i + ([p_3, l_1] + l_3, p_1)j$

$(i2\bar{h}p_3)k + (i2\bar{h}p_1)i + (i2\bar{h}p_2)j = i2\bar{h}\boldsymbol{p}$

proof b:

$i\bar{h}(p \times l - l \times p) = (p_1l_2 - p_2l_1 - l_1p_2 + l_2p_1)k + (p_2l_3 - p_3l_2 - l_2p_3 + l_3p_2)i + (p_3l_1 - p_1l_3 - l_3p_1 + l_1p_3)j$

$[l^2, p] = [l^2, p] = [l_x^2 + l_y^2 + l_z^2, p]$

$[l_x^2, p] = l_x[l_x, p] + [l_x, p]l_x = l_x(i\bar{h}(p_zj - p_yk)) + (i\bar{h}(p_zj - p_yk))l_x = i\bar{h}(l_xp_z + p_zl_x)j - i\bar{h}(l_xp_y + p_yl_x)k$

$[l_y^2, p] = l_y[l_y, p] + [l_y, p]l_y = l_y(i\bar{h}(p_xk - p_zi)) + (i\bar{h}(p_xk - p_zi))l_y = i\bar{h}(l_yp_x + p_xl_y)k - i\bar{h}(l_yp_z + p_zl_y)i$

$[l_z^2, p] = l_z[l_z, p] + [l_z, p]l_z = l_zi\bar{h}(p_yi - p_xj) + i\bar{h}(p_yi - p_xj)l_z = i\bar{h}(l_zp_y + p_yl_z)i - i\bar{h}(l_zp_x + p_xl_z)j$

$[l^2, p] = i\bar{h}(l_yp_x + p_xl_y - l_xp_y - p_yl_x)k + i\bar{h}(l_zp_y + p_yl_z - l_yp_z - p_zl_y)i + i\bar{h}(l_xp_z + p_zl_x - l_zp_x - p_xl_z)j$

so we have $i\bar{h}(p \times l - l \times p) = [l^2, p]$

4 Oct 30 HW

$\psi_n = A_n e^{-\alpha^2x^2/2}H_n(\alpha x)$

2.7

(1)

$H_n(\alpha x) = \frac{1}{2\alpha x}(H_{n+1}(\alpha x) + 2nH_{n-1}(\alpha x))$

$x\psi_n = \frac{1}{2\alpha}(H_{n+1}(\alpha x) + 2nH_{n-1}(\alpha x))A_ne^{-\alpha^2x^2/2}$

$A_n = \frac{1}{\sqrt{2n}}A_{n-1}; A_n = \sqrt{2n+2}A_{n+1}$

so we get:$x\psi_n = \frac{1}{2\alpha}(A_nH_{n+1}(\alpha x) + 2nA_nH_{n-1}\alpha x)e^{-\alpha^2x^2/2}$

$\frac{1}{2\alpha}(\sqrt{2(n+1)}A_{n+1}(\alpha x) + 2n\frac{1}{\sqrt{2n}}A_{n-1}H_{n-1}\alpha x)e^{-\alpha^2x^2/2}$

$= \frac{1}{2\alpha}(\sqrt{2(n+1)}\psi_{n+1} + \sqrt{2n}\psi_{n-1})$

(2)

$x^2\psi_n = \frac{1}{2\alpha}(\sqrt{2(n+1)}x\psi_{n+1} + \sqrt{2n}x\psi_{n-1})$

$x^2\psi_n = \frac{1}{2\alpha}(\sqrt{2(n+1)}[\frac{1}{2\alpha}(\sqrt{2(n+2)}\psi_{n+2} + \sqrt{2(n+1)}\psi_n)] + \sqrt{2n}[\frac{1}{2\alpha}(\sqrt{2n}\psi_n + \sqrt{2(n-1)}\psi_{n-2})]) = \frac{1}{4\alpha^2}[\sqrt{(4(n+1)(n+2))}\psi_{n+2} + 2(n+1)\psi_n + 2n\psi_n + \sqrt{4n(n-1)}\psi_{n-2}]$

$= \frac{1}{2\alpha^2}[\sqrt{(n+1)(n+2)}\psi_{n+2} + (2n+1)\psi_n + \sqrt{n(n-1)}\psi_{n-2}]$

(3)

$\bar{x} = \int \psi_n^* x\psi_n = 0$

$V = \frac{1}{2}Kx^2; \bar{V} = \int \psi_n^* V\psi_n = \frac{1}{2}K\int \psi_n^* x^2\psi_n = \frac{1}{2}K\frac{1}{2\alpha^2}(2n+1) = \frac{K(2n+1)}{4\alpha^2} = \frac{\omega(2n+1)\bar{h}}{4} = \frac{E_n}{2}$

2.8

$H_n'(\alpha z) = 2n\alpha H_{n-1}(\alpha z)$

$\psi_n' = A_n e^{-\alpha^2x^2/2} 2n\alpha H_{n-1}(\alpha x) - \alpha^2 x A_n e^{-\alpha^2x^2/2}[\frac{1}{2\alpha x}(H_{n+1} + 2nH_{n-1})]$
$=2n\alpha A_n e^{-\alpha^2x^2/2} H_{n-1} - \frac{\alpha}{2} A_n e^{-\alpha^2x^2/2} H_{n+1} - \alpha n A_n e^{-\alpha^2x^2/2} H_{n-1}$
$=\alpha n A_n e^{-\alpha^2x^2/2} H_{n-1} - \frac{\alpha}{2} A_n e^{-\alpha^2x^2/2} H_{n+1}$
$A_n = \frac{1}{\sqrt{2n}} A_{n-1}; A_n = \sqrt{2n+2} A_{n+1}$
$=\alpha\sqrt{\frac{n}{2}}\psi_{n-1} - \alpha\sqrt{\frac{n+1}{2}}\psi_{n+1}$
(2)
$\frac{d}{dx}\psi_n' = \alpha[\sqrt{\frac{n}{2}}\psi_{n-1}' - \sqrt{\frac{n+1}{2}}\psi_{n+1}']$
$\frac{d}{dx}\psi_n' = \alpha[\sqrt{\frac{n}{2}}[\alpha\sqrt{\frac{n-1}{2}}\psi_{n-2} - \alpha\sqrt{\frac{n}{2}}\psi_n] - \sqrt{\frac{n+1}{2}}[\alpha\sqrt{\frac{n+1}{2}}\psi_n] - \alpha\sqrt{\frac{n+2}{2}}\psi_{n+2}]$
$\frac{d}{dx}\psi_n' = \alpha^2/2[\sqrt{(n-1)n}\psi_{n-2} - n\psi_n - (n+1)\psi_n + \sqrt{(n+1)(n+2)}\psi_{n+2}]$
$\frac{d}{dx}\psi_n' = \alpha^2/2[\sqrt{(n-1)n}\psi_{n-2} - (2n+1)\psi_n + \sqrt{(n+1)(n+2)}\psi_{n+2}]$
(3)
$\bar{p} = \int \psi_n^* p \psi_n = 0$
$\bar{T} = \int \psi_n^* T \psi_n = \int \psi_n^* (-\bar{h}^2/(2m))\frac{d^2}{dx^2}\psi_n = \frac{-\bar{h}^2}{2m}\int \psi_n^* \frac{d^2}{dx^2}\psi_n = \frac{-\bar{h}^2}{2m}[-\frac{\alpha^2}{2}(2n+1)] = \frac{\bar{h}\alpha^2}{4m}(2n+1) = \frac{\bar{h}\omega}{4}(2n+1) = \frac{1}{2}E_n$
2.9
$\Delta x \Delta p = \sqrt{\bar{x^2}\bar{p^2}} = \sqrt{E_n m E_n/K} = \frac{E_n}{\omega} = \frac{\bar{h}}{2}(2n+1)$
3.14
l_zEigenstate:$\psi_m = \sqrt{\frac{1}{2\pi}} e^{im\phi}$
$\int \phi^* l_z \phi = \int (l_z\phi)^*\phi$
$\int \phi^*[l_y, l_z]\phi = \int \phi^* l_y l_z \phi - \phi^* l_z l_y \phi = \int \phi^* l_y \phi \bar{h} m - (l_z\phi)^* l_y \phi = \bar{h} m \bar{l}_y - \bar{h} m \bar{l}_y = 0 = i\bar{h} l_x$
$\int \phi^*[l_z, l_x]\phi = \int \phi^* l_z l_x \phi - \phi^* l_x l_z \phi = \int (l_z\phi)^* l_x \phi - (l_x\phi)^* l_z \phi = \int \phi^* \bar{h} m l_x \phi - \bar{h} m \phi^* l_x \phi = 0 = i\bar{h} l_y$

5 Nov 4 HW

3.15
$\psi = Y_{lm}$ so $\bar{l_x^2} = \int \psi^* l_x^2 \psi d\Omega$ and $l_x^2 = \frac{1}{i\bar{h}}[l_y, l_z]l_x = \frac{1}{i\bar{h}}(l_y l_z - l_z l_y)l_x = \frac{1}{i\bar{h}}(l_y l_z l_x - l_z l_y l_x)$ and we know $l_z l_x = l_x l_z + i\bar{h} l_y$ so :$\frac{1}{i\bar{h}}(l_y l_x l_z + i\bar{h} l_y^2 - l_z l_y l_x) = \frac{1}{i\bar{h}} l_y l_x l_z + l_y^2 - \frac{1}{i\bar{h}} l_z l_y l_x = l_x^2$
so $\bar{l_x^2} = \int \psi^*(\frac{1}{i\bar{h}} l_y l_x l_z + l_y^2 - \frac{1}{i\bar{h}} l_z l_y l_x)\psi d\Omega = \bar{l_y^2} + \int \psi^* \frac{1}{i\bar{h}} l_t l_x \bar{h} m \psi - \frac{1}{i\bar{h}} \bar{h} m \psi^* l_y l_x \psi d\Omega = \bar{l_y^2}$
so we have $\bar{l_x^2} = \bar{l_y^2}$ now
we know $l^2 = l_x^2 + l_y^2 + l_z^2$,so $\bar{l^2} = 2\bar{l_x^2} + \bar{l_z^2}$
$\bar{l_x^2} = \frac{1}{2}(\int \psi^* l^2 \psi - \psi^* l_z^2 \psi d\Omega) = \frac{1}{2}(l(l+1)\bar{h}^2 - m^2\bar{h}^2)$
so $(\Delta \bar{l}_x)^2$ we have :$\Delta l_x = \sqrt{-\bar{l_x}^2 + \bar{l_x^2}} = \sqrt{\bar{l_x^2}}$ and thus $(\Delta l_x)^2 = |\bar{l_x^2}| = \frac{1}{2}(l(l+1)\bar{h}^2 - m^2\bar{h}^2)$ so we have
$(\Delta \bar{l}_x)^2 = \frac{1}{2}(l(l+1)\bar{h}^2 - m^2\bar{h}^2) = (\Delta \bar{l}_y)^2$
3.16
set $\psi = c_1 Y_{11} + c_2 Y_{20}$

$l_z\psi = c_1\bar{h}Y_{11} + c_2 \times 0 = c_1\bar{h}Y_{11}$ so the posibility of status $0\bar{h}$ is $|c_2|^2$,and $\bar{h}$ is 1.so $l_z = \bar{h}$ and the average $=|c_1|^2\bar{h}$

$l^2\psi = c_1 \times 1 \times 2Y_{11}\bar{h}^2 + c_2 \times 2 \times 3Y_{20}\bar{h}^2$ so l^2 can be $2\bar{h}^2$posibility is $|c_1|^2$,and $6\bar{h}^2$ posibility is $|c_2|^2$.

6 Nov 6 HW

*3.18

$e^{-aA}Be^{aA} = \Sigma\frac{-a^nA^n}{n!}B\Sigma\frac{a^mA^m}{m!} = \Sigma\Sigma\frac{-a^nA^n}{n!}B\frac{a^mA^m}{m!} = \Sigma_{m,n}\frac{-1^na^{m+n}A^nBA^m}{n!m!}$

$\Sigma_{m,n}\frac{-1^na^{m+n}A^nBA^m}{n!m!} = B - aAB + aBA - a^2ABA + a^2BA^2/2 + a^2A^2B/2...$

$=B - a[A,B] - \frac{1}{2}a^2(A[A,B])...$ for any m+n=i we can have

$a^i\Sigma_{m+n=i}\frac{-1^nA^nBA^m}{n!m!}$ for i=3 we can have:$a^3\Sigma_{m+n=3}\frac{-1^nA^nBA^m}{n!m!}$

$= a^3[[A,[A,B]],A]/6 = -\frac{a^3}{3!}[A,[A,[A,B]]]$

Now set for i = 0,1,2,3 we set $F_i = \frac{-a}{i+1}[A,F_{i-1}] = \Sigma_{m+n=i}\frac{a^i(-1)^nA^nBA^m}{m!n!}$

Now consider for ant k satisfied F_k satisfied :$F_k = \frac{a^k}{k!}(-1)^k[A,[A,[A,...[A,B]]]] = \Sigma_{m+n=k}\frac{a^k(-1)^nA^nBA^m}{m!n!}$ and for k+1 we have:

$-a/(k+1)[A,F_k] = \frac{a^{k+1}}{(k+1)!}(-1)^{k+1}[A,[A,...[A,B]]] = \Sigma_{m+n=k}\frac{a^{k+1}(-1)^n[A,A^nBA^m]}{m!n!(k+1)}$

and for $[A,A^nBA^m] = [A,A^n]BA^m + A^n[A,BA^m] = [A,A^n]BA^m + A^n[A,B]A^m + A^nB[A,A^m] = A^n[A,B]A^m = A^{n+1}BA^m - A^nBA^{m+1} = A^{n+1}BA^m + (-1)A^nBA^{m+1}$so

$\Sigma_{m+n=k}\frac{a^{k+1}(-1)^n(A^{n+1}BA^m + (-1)A^nBA^{m+1})}{m!n!(k+1)}$

$\Sigma_{m+n=k}\frac{a^{k+1}(-1)^{n+1}A^{n+1}BA^m + a^{k+1}(-1)^nA^nBA^{m+1}}{m!n!(k+1)} = \Sigma_{m+n'=k+1}\frac{a^{k+1}(-1)^{n'}A^{n'}BA^m}{m!(n'-1)!(k+1)} + \Sigma_{m'+n=k+1}\frac{a^{k+1}(-1)^nA^nBA^{m'}}{(m'-1)!n!(k+1)} - \frac{-a^{k+1}ABA^k}{(k+1)!} - \frac{a^{k+1}(-1)^kA^kBA}{(k+1)!} = \Sigma_{m+n'=k+1}n'\frac{a^{k+1}(-1)^{n'}A^{n'}BA^m}{m!(n')!(k+1)} + \Sigma_{m'+n=k+1}m'\frac{a^{k+1}(-1)^nA^nBA^{m'}}{(m')!n!(k+1)} + \frac{a^{k+1}ABA^k}{(k+1)!} - \frac{a^{k+1}(-1)^kA^kBA}{(k+1)!}$ 1 and 3 combine can have a m$\neq$ 1term and 2 and 4 combine can have a $n \neq 1$ term :$\Sigma_{m+n'=k+1;m\neq1}n'\frac{a^{k+1}(-1)^{n'}A^{n'}BA^m}{m!(n')!(k+1)} + \Sigma_{m'+n=k+1;n\neq1}m'\frac{a^{k+1}(-1)^nA^nBA^{m'}}{(m')!n!(k+1)}$ so we get satisfied for full k+1 terms : m=0,1,2...,k total k+1,and we set m+n=k+1,we get:$\Sigma\frac{a^{k+1}(-1)^nA^nBA^m}{m!n!}$

PROOF DONE.

7 Nov 13

4.4: proof:$-\hbar^2\frac{d^2}{dt^2}\bar{A} = \overline{[[A,H],H]}$

PROOF:$\frac{d}{dt}\bar{C} = \frac{1}{i\hbar}\overline{[C,B]}$ and let $\bar{C} = \frac{d}{dt}\bar{A} = \frac{1}{i\hbar}\overline{[A,H]}$ so

$\frac{d}{dt}\frac{d}{dt}\bar{A} = -\frac{1}{\hbar^2}\overline{[[A,H],H]}$

PROVED

4.5: proof:$\frac{d}{dt}\bar{A} = 0$ $\bar{A}$is under a stationary state.

PROVE:$\frac{d}{dt}\bar{A} = \frac{1}{i\hbar}\overline{[A,H]} = \frac{1}{i\hbar}\int\psi^*AH\psi - \psi^*HA\psi = \frac{1}{i\hbar}(E\int\psi^*A\psi - E\psi^*A\psi) = 0$

DONE

8 Nov 18

homework:

4.2:

(a)

$$H = h(q_1) + h(q_2)$$
$$q_1, q_2 = \phi_1, \phi_1; \phi_1, \phi_2; \phi_1, \phi_3; \phi_2, \phi_2; \phi_2, \phi_3; \phi_3, \phi_3.$$

6 kinds

(b)

$$H = h(q_1) + h(q_2)$$
$$q_1, q_2 = \phi_1, \phi_2, \phi_2, \phi_1; \phi_1, \phi_3; \phi_3, \phi_1; \phi_2, \phi_3; \phi_3, \phi_2.$$

3 kinds

(c)

$$3 \times 3 = 9$$

9 kinds

4.3:

(a)

$$3^3 = 27$$

27 kinds

(b)

$$\phi_{q_1}, \phi_{q_2}, \phi_{q_3}:$$
$$1, 2, 3$$

1 kind

(c)

$$\phi_{q_1}, \phi_{q_2}, \phi_{q_3}:$$
$$1, 1, 1; 1, 1, 2; 1, 1, 3; 1, 2, 2; 1, 2, 3;$$
$$1, 3, 3; 2, 2, 2; 2, 2, 3; 2, 3, 3; 3, 3, 3;$$

10 kinds

4.8:

proof:

$$(\frac{dF}{dt})_{nn'} = i\omega_{nn'}F_{nn'}, \omega_{nn'} = (E_n - E_{n'})/\hbar \tag{1}$$

prove:

$$\frac{dF}{dt} = \frac{1}{i\hbar}[F(t), H]$$

$$(\frac{dF}{dt})_{nn'} = \frac{1}{i\hbar}\int \phi_n FH\phi_{n'} - \phi_n HF\phi_{n'}$$

$$= \frac{1}{i\hbar}(E_{n'}\int \phi_n F\phi_{n'} - E_n\phi_n F\phi_{n'})$$

$$= \frac{1}{i\hbar}(E_{n'} - E_n)\int \phi_n F\phi_{n'}$$

$$= -\frac{1}{i\hbar}\hbar\omega_{nn'}F_{nn'}$$

$$= i\omega_{nn'}F_{nn'}$$

9 Nov 20

4.7: proof:

$$\frac{\partial E_n}{\partial \lambda} =< \psi_n | \frac{\partial H}{\partial \lambda} | \psi_n > \tag{2}$$

PROVE:

$$\begin{aligned}
&E_n \psi_n = \hat{H} \psi_n \\
&\psi_{na}^* E_n \psi_n^a = E_n = \psi_{na}^* \hat{H} \psi_n^a \\
&\frac{\partial E_n}{\partial \lambda} = \frac{\partial}{\partial \lambda} (\psi_{na}^* \hat{H} \psi_n^a) \\
&= \psi_{na}^* \frac{\partial \hat{H}}{\partial \lambda} \psi_n^a =< \psi_n | \frac{\partial \hat{H}}{\partial \lambda} | \psi_n >
\end{aligned}$$

PROVED